

Due -April 19th

General Instructions:

Form a group of up to 4.

Choose one group member's "Easy" and "Hard" week training and test sets from Deliverable 2.

All forecasts will be done at 60 minute intervals.

All results will be communicated in a power point presentation. Record a zoom session reviewing your slides. Your presentation should be at least 7 minutes and no more than 15.

	Learning Objectives	Deliverables	
1	Students can use ACF and PACF plots to identify possible orders of an ARIMA (p,d,q) x (P,D,Q)s	*Create ACF and PACF plots for your deseasonalized and detrended data. (2 pts) *Comment on the combination of the ACF Plot and PACF plot and identify if it appears as though an MA(q) process and/or an AR(p) process may be involved for both (p,d,q) and (P,D,Q)s processes. (3 pt -comments) *What order process may be involved? Describe what ARIMA models you want to try and why. (2 pt)	7 pts
2	Student can fit a seasonal ARIMA (p,d,q)x(P,D,Q)s model to their data.	Do the following for each training period (NOTE: FITTING ARIMA MODELS WITH LARGE SEASONS CAN BE VERY TIME CONSUMING) : *Fit each of the different model parameter options to your data and record the fit information (AIC etc) (minimum 4 choices)(4pt) *Additionally use the auto.arima() function with specified 168 hr season and record the provided (p,d,q)x(P,D,Q) orders *Provide the residual plots of your 'best' model (4 in 1 residual plots, ACF of Residuals) and discuss them. (4pt) *Plot the fit of your 'best' fitting model (4pt) *Comment on the fit of your best model. (4pt)	16 pts
3	Student can use a fitted ARIMA (p,d,q)x(P,D,Q)s model to create a Forecast	Do the following for each test period: *Using several competing models from step 2, evaluate their forecast performance on the 'Easy Week' and the 'Hard Week' (2pt) *Suggest an overall best model based on fit and forecast performance (2pt) *Graph your favorite forecast for each week including confidence intervals (2pt) *Comment on the forecast performance of your overall best model (2pt)	8 pts

4	Students can use dynamic regression to model time series data. Students can identify variables that are important to the models. Students can identify the lag at which variables may be relevant.	Sections 4 and 5 can be done using either your easy or hard week. (DO NOT DO BOTH WEEKS) Note: Here it may be helpful to increase the size of your training set to get a wider range of values for your regressors in the training set. 1. Identify a subset of the provided features which you think may be relevant for predicting your time series (potential regressors) Creative plotting may be helpful here. (1pt) 2. Plot your regressors along with your time series data. Comment on any relationships you observe. If you have many many regressors, this may be a good opportunity to remove some. (2pt) 3. Fit a dynamic regression model using your regressors. You may use auto.arima. Do not use any lagged regressors yet. (1pt) 4. What is the error (model fit) of your dynamic regression model? (1pt) 5. Try using more or less regressors, can you find a better model? Provide evidence of testing multiple options, such as a table capturing models tried and their associated fits. (3pt) 6. Provide a graph of the best fit model and the residuals(4 pt) 7. Comment on the fit of the model. Do the regressors appear to be helping compared to your model without regressors? (3pt)	15 pts
5		1. Investigate if using lagged regressors can improve your model. Use a cross correlation plot to identify if lagged values may be helpful. (2pt) 2. Report your findings including model fits, and different lags tested. (2pt) 3. Compare your lagged model performance to your no lagged data model performance. (1pt) 4. Provide a graph of the best fit lagged model. If no lags were helpful, there is no need to redraw the graph from before. (1pt) 5. Comment on the fit of the model and residuals. Do the lagged regressors appear to be helping? (3pt)	9 pts
6	Students can use a dynamic regression model to make forecasts.	1. Forecast your Easy and Hard weeks using your Best Dynamic Regression Model from steps 4 and 5. Note you will need to retrain your model up to the start of your easy and hard weeks before forecasting.(4pt) 2. Choose a time period that you would expect to showcase your dynamic regression model performance. Train (using previously identified model) and Forecast for the period. How did it do? Did it beat winters and ARIMA? (For example, if your regressors include snowfall, and you expect that regressor to help predict low periods of solar production, find a period of time to forecast where there was snow. Note that snow will also need to be present in the training data for the model to learn about snow!) (1pt) 3. Provide forecast performance measures for all models and time periods tested. (3pt) 4. Provide graphs of your forecasts (Easy, Hard, and Showcase Week).(3pt) 5. Comment on the fit of your forecasts and residuals of your forecasts. Be insightful with your observations. (3pt) 6. Comment on which additional regressors you might still like to try? Comment on which regressors were and weren't helpful of the ones you chose to try first.(1pt)	15 pts

7	Students can capture complex seasonality using Fourier Transforms and indicator variables.	1. Use Fourier Transforms to capture the different seasonalities of your data and feed the fourier transform data as regressors to a dynamic regression model. (Train and forecast using this model + other regressors you found helpful from before for your EASY and Hard weeks, you may use auto arima to find the best model parameters) (4pt) 2. Compare the fourier transform model to a model that uses indicators variables to capture seasonality. Which model performs better? Evaluate the residuals of both models. (4pt) 3. Add these models to your "table of all model performance" (2pt)	10 pts
8	Student can compare and choose between different forecasting models	1. Create a table that summarizes all models so far and their performance for the easy and hard weeks. The table should include Exponential Smoothing, seasonal ARIMA, dynamic regression, dynamic regression with fourier transforms capturing seasonality, dynamic regression with indicator variables capturing seasonality (2pt) 2. Discuss which model you would recommend and why (8pts)	10 pts
9	Students can explain their modeling approach and defend their modeling decisions to an audience.	1. Submit a power point presentation to document the work completed. 2. Be prepared to give a presentation in class. The presentation should be between 10-20 minutes in length. Each team member should talk during the presentation.	30 pts
10		3. Turn in your R code used along with any necessary input files.	15 pts